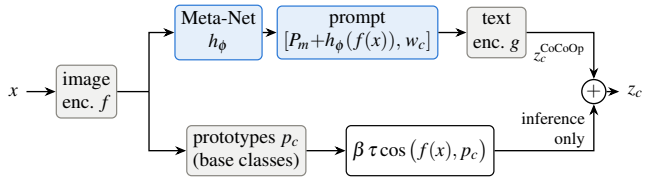
ProtoCoCoOp: Knowledge Distillation and Visual Prototypes for Base-to-Novel Generalization of CLIP

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https://github.com/giampyhatesyou/CLIP_Project-UniTn

Abstract. Prompt learning adapts CLIP to a new task from a handful of labeled images, but the learned prompts overfit the classes seen in training and erode the zero-shot transfer that made CLIP useful in the first place. We study this base-to-novel trade-off on Flowers102 with 10 shots per class and propose ProtoCoCoOp, which adds two light components to CoCoOp: during training, a distillation term keeps the adapted model close to the frozen CLIP teacher; at inference, a training-free prototype residual adds visual evidence to the logits of the classes seen in training. With a frozen ViT-B/16 backbone and 37K trainable parameters, ProtoCoCoOp reaches a harmonic mean of 84.44%, against 74.62% for zero-shot CLIP and 83.38% for CoCoOp. The ablation shows why: distillation moves accuracy from base to novel classes, and prototypes give the base accuracy back without touching novel predictions.



Training (10 images per base class): $\mathscr{L} = (1-\alpha)\,\mathscr{L}_{\text{CE}} + \alpha\,T^2\,\text{KL}\big(p^{\text{CLIP}} \,\|\, p^{\text{student}}\big)$, teacher: frozen CLIP with "a photo of a {class}"

Figure 1: ProtoCoCoOp. Blue blocks are trained (37K parameters), grey blocks are frozen CLIP. Distillation acts only during training; the prototype residual only at inference and only on base classes.

1 Introduction

Vision–language models such as CLIP [1] embed images and text in a shared space and can recognize arbitrary categories by comparing an image with prompts such as "a photo of a {class}". When a few labeled examples of the target task are available, prompt learning methods such as CoOp [2] replace the hand-written prompt with learned context vectors and improve accuracy substantially. The price is generalization: prompts optimized on a set of *base* classes transfer poorly to *novel* classes of the same task that were never seen in training [3]. CoCoOp [3] reduces the problem by conditioning the prompt on each image, but a gap remains.

This report studies the base-to-novel setting on Flowers102 [4] and asks whether the two sides of the trade-off can be handled separately. ProtoCoCoOp (Figure 1) keeps CoCoOp as the backbone and adds:

- a **knowledge-distillation** term [5] that pulls the student towards the zero-shot predictions of the frozen CLIP teacher during training, protecting novel-class accuracy;
- a **prototype residual** applied only at inference, which adds the similarity between the test image and per-class visual prototypes [6, 7] to the logits of base classes, recovering the base accuracy that distillation gives up.

Neither component adds trainable parameters, and the residual needs no training at all. Among the configurations we compare, the combination obtains the best harmonic mean between base and novel accuracy (Table 1).

2 Method

2.1 Prompt learning with CoCoOp

CLIP encodes an image x into $f(x)$ and a text t into $g(t)$, both L2-normalized, and scores class c with the scaled cosine similarity $z_c = \tau\,\langle f(x), g(t_c)\rangle$, where τ is CLIP's learned logit scale. CoOp [2] replaces the words of a hand-written prompt with M learnable context vectors, $t_c = [P_1, \ldots, P_M, w_c]$, where w_c is the embedding of the class name. CoCoOp [3] adds a small Meta-Net h_ϕ that maps the image feature to a bias shared by all context tokens,

$$P_m(x) = P_m + h_\phi\big(f(x)\big), \qquad m = 1, \ldots, M, \tag{1}$$

so that the prompt adapts to each instance. Only $\{P_m\}$ and ϕ are trained, with cross-entropy on the base classes; both encoders stay frozen.

2.2 Distillation from the frozen teacher

Let z^s be the student logits and z^t the logits of the same frozen CLIP with the hand-written prompt "a photo of a {class}". Following Hinton et al. [5], we minimize

$$\mathscr{L} = (1-\alpha)\,\mathscr{L}_{\text{CE}} + \alpha\,T^2\,\text{KL}\big(\sigma(z^t/T) \,\|\, \sigma(z^s/T)\big), \tag{2}$$

with $\alpha = 0.3$ and $T = 2$, where σ is the softmax. The cross-entropy term only sees base classes, while the KL term runs over the whole label vocabulary, so it also constrains how the learned context treats class names that never receive a label. The teacher acts as a soft anchor: the student may specialize only as far as it stays consistent with CLIP's zero-shot geometry. KgCoOp [8] and PromptSRC [9] use related regularizers towards the frozen model.

2.3 Prototype residual at inference

For each base class c we encode its 10 training images, plus 10 augmented views of each (random crop, horizontal flip, rotation up to 30°, color jitter), with the frozen image encoder, and average the 110 normalized embeddings into a prototype p_c, normalized again to unit length. At test time the base-class logits become

$$z_c(x) = z_c^{\text{CoCoOp}}(x) + \beta\,\tau\cos\big(f(x), p_c\big), \qquad c \in \mathscr{C}_{\text{base}}, \tag{3}$$

with $\beta = 0.5$; novel classes have no prototype and keep their CoCoOp logits. The residual is related to the cache model of Tip-Adapter [7], but it uses one mean prototype per class and is added to the prompt-based classifier instead of replacing it. It costs one 512×51 matrix product per image and no gradient step.

3 Experimental setup

Data and protocol. Flowers102 [4] contains 102 flower species. Following the base-to-novel protocol of [3], the first

Method	Base	Novel	HM
CLIP zero-shot [1]	71.33	**78.24**	74.62
CoCoOp [3]	93.57	75.19	83.38
CoCoOp + KD	87.18	76.47	81.47
ProtoCoCoOp (ours)	**93.85**	76.74	**84.44**

Table 1: Test accuracy (%) on Flowers102, base-to-novel, 10 shots per base class, ViT-B/16. ProtoCoCoOp is CoCoOp trained with KD and evaluated with the prototype residual.

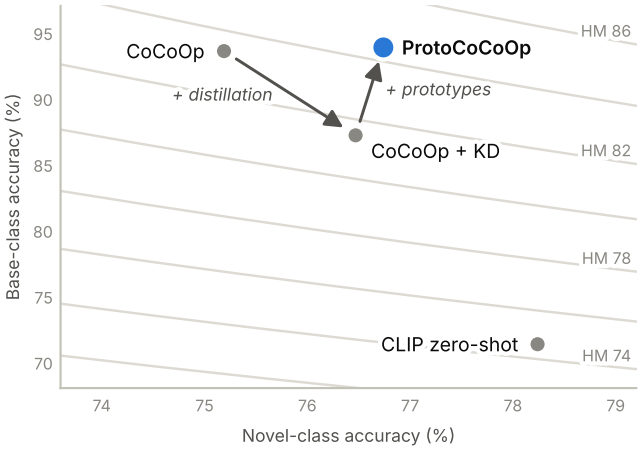


Figure 2: Base vs. novel accuracy. Grey curves join points with equal harmonic mean. Distillation moves CoCoOp down and to the right; prototypes then move it straight up.

51 classes are base and the other 51 novel. We train on the 10 images per base class of the official training split (510 images), select the checkpoint on the base classes of the validation split, and test on the official test split. Base and novel accuracy are each measured as 51-way classification among the classes of their split, and we report their harmonic mean $\mathrm{HM} = 2BN/(B+N)$, which rewards balanced performance.

Implementation. CLIP ViT-B/16, frozen and run in fp32. $M = 8$ context vectors drawn from $\mathcal{N}(0, 0.02^2)$; Meta-Net $512 \to 32 \to 512$ with ReLU; 37,408 trainable parameters, about 0.025% of CLIP. SGD with learning rate 0.002, momentum 0.9, weight decay $5 \cdot 10^{-4}$, batch size 1 and a cosine schedule over at most 15 epochs, with early stopping (patience 5) on base validation accuracy; seed 42. All configurations share these hyper-parameters. Zero-shot CLIP uses the prompt "a photo of a {class}, a type of flower."

4 Results

Prompt learning buys base accuracy with novel accuracy. CoCoOp improves base accuracy by 22.2 points over zero-shot CLIP and loses 3.1 points on novel classes; HM still rises to 83.38 because the base gain is much larger. Our zero-shot numbers (71.33 / 78.24) are close to the 72.08 / 77.80 reported in [3] on the same split, a useful check of the evaluation pipeline.

Distillation moves the operating point rather than improving it. Adding KD recovers 1.3 novel points but costs 6.4 base points, so HM drops to 81.47. On its own, distillation trades one side of the problem for the other.

Prototypes give the base accuracy back. CoCoOp + KD and ProtoCoCoOp use the same training objective and differ only at inference, so the base gain from 87.18 to 93.85 (+6.7) comes from the prototype residual. Prototypes cannot affect novel predictions; the 0.27-point difference in novel accuracy between the two rows is variation between their training runs. The final model matches CoCoOp on base classes (93.85 vs. 93.57) while keeping 1.55 more points on novel ones, for the best HM: 84.44, that is +1.06 over CoCoOp and +9.82 over zero-shot CLIP.

5 Discussion

What the experiments show. On this benchmark the trade-off of prompt learning can be split into two problems handled at different times: a training-time regularizer that preserves CLIP's zero-shot behavior, and an inference-time memory that restores specialization where labeled data exist. The two act on different sides of the trade-off—distillation on the text side and on all classes, prototypes on the visual side and on base classes only—which is why their effects add up in HM instead of cancelling out.

What they do not show. The evidence is narrower than that conclusion. (i) Every number comes from one training run (seed 42) on one dataset; a gain of about one HM point over CoCoOp should be confirmed over several seeds and the other datasets of [3]. (ii) The prototype-only configuration (CoCoOp + prototypes without KD) is supported by the code but not reported, so the ablation does not establish that distillation is necessary in the final model. (iii) The residual is safe only because base and novel classes are evaluated separately; in a joint 102-way decision a positive term on base classes alone would bias predictions towards them and would need calibration. (iv) Distillation sees the names of novel classes, though never their images or labels, which is slightly weaker than a strictly inductive protocol. (v) No sensitivity analysis of α, β and T is reported. Novel accuracy also stays 1.5 points below zero-shot CLIP: the method narrows the generalization gap rather than closing it.

6 Conclusion

ProtoCoCoOp shows that a few-shot adapted CLIP can keep its specialization while giving up less of its zero-shot generalization, using two components that add no trainable parameters: distillation during training and prototype residuals at inference. Natural next steps are a multi-seed evaluation on the eleven datasets of the CoCoOp benchmark, the missing prototype-only ablation, a joint base-and-novel evaluation with a calibrated residual, and a teacher restricted to base-class names and prompted with the stronger dataset-specific template.

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